

Problem set 1

Due Thursday January 22, 11:59pm

Problem sets will be assigned weekly. You should submit your solutions to the problem sets in PDF form via the Gradescope interface in Canvas. LaTeX is a very nice tool to use for this purpose, but handwritten solutions are good too as long as they are clearly legible.

On the problem sets, you are free — and indeed strongly encouraged — to collaborate with other students in our class. Discussing mathematics with others is the easiest way of learning it.

There are two rules:

- (1) *You are not permitted to post problems from the problem sets to publicly accessible websites.*
- (2) *You must write down your solutions to the problems on your own: thus your solutions should not be identical or near-identical to solutions submitted by other students, and they also should not look like output from ChatGPT or similar tools.*

You are of course free and encouraged to discuss the problems with me or with our TA or ULAs. You are also free to use other resources, such as our textbook, our lecture notes, any other books, information found on the Internet, or AI tools. (Despite this last point, I do not suggest searching the Internet for solutions to the problems, or using AI to solve the problems; even if you do get correct solutions, you won't learn much that way.)

In writing up your solutions, you are free to use without proof anything stated in class or in the lecture notes, or anything in the course text. You should be sure to state clearly what you are using. If you have some doubt about whether something is OK to use, just ask!

Exercise 1.1 (5 points). Suppose A , B , and C are sets, $f : A \rightarrow B$ is surjective, and $g : B \rightarrow C$ is surjective. Show that $g \circ f$ is surjective.

Exercise 1.2 (10 points; Rudin 1.3). Suppose F is a field with $x, y, z \in F$. Prove carefully from the field axioms:

- (1) If $x \neq 0$ and $xy = xz$, then $y = z$.
- (2) If $x \neq 0$ and $xy = x$, then $y = 1$.
- (3) If $xy = 1$, then $x \neq 0$ and $y = x^{-1}$.
- (4) If $x \neq 0$, then $(x^{-1})^{-1} = x$.

Exercise 1.3 (10 points). In this problem you may freely use without proof all the standard facts about arithmetic of rational numbers and integers.

- (1) Prove that $\mathbb{Q}$ does not contain any x with $x^2 = 3$. (For this part it may be useful to use the fundamental theorem of arithmetic, which you may use without proof.)
- (2) Let $\mathbb{Q}(\sqrt{3})$ be the set of ordered pairs (a, b) with $a, b \in \mathbb{Q}$. Informally, we think of each pair (a, b) as representing a number " $a + b\sqrt{3}$." We can equip $\mathbb{Q}(\sqrt{3})$ with addition and product laws as follows:

$$\begin{aligned}(a, b) + (a', b') &= (a + a', b + b'), \\ (a, b) \cdot (a', b') &= (aa' + 3bb', ab' + ba').\end{aligned}$$

(These rules are motivated by applying the usual rules of arithmetic to sums of the form " $a + b\sqrt{3}$ ".)

Show that $\mathbb{Q}(\sqrt{3})$ can be made into a field, with these addition and product laws. (This means saying carefully what are the 0 and 1 elements, what is the negation law, what is the inversion law, and then proving that all the axioms of a field are satisfied. This will probably be the lengthiest part of your writeup.)

- (3) In solving part (2), did you use the result of part (1)? If so, where?
- (4) $\mathbb{Q}(\sqrt{3})$ contains an element 3, defined as $3 = 1 + 1 + 1$, where 1 is the 1 element you already defined in part (2). What is the element 3? Prove that $\mathbb{Q}(\sqrt{3})$ contains exactly two elements x with $x^2 = 3$.

Exercise 1.4 (8 points). Let A be a nonempty subset of an ordered set S . Suppose α is a lower bound of A and β is an upper bound of A .

- (1) Prove that $\alpha \leq \beta$.
- (2) Note that in the statement of the problem we require A to be nonempty. Would the statement $\alpha \leq \beta$ still hold even if we allow A to be empty? Explain why or why not. In solving part (1), did you use the fact that A is nonempty? If so, where did you use it?